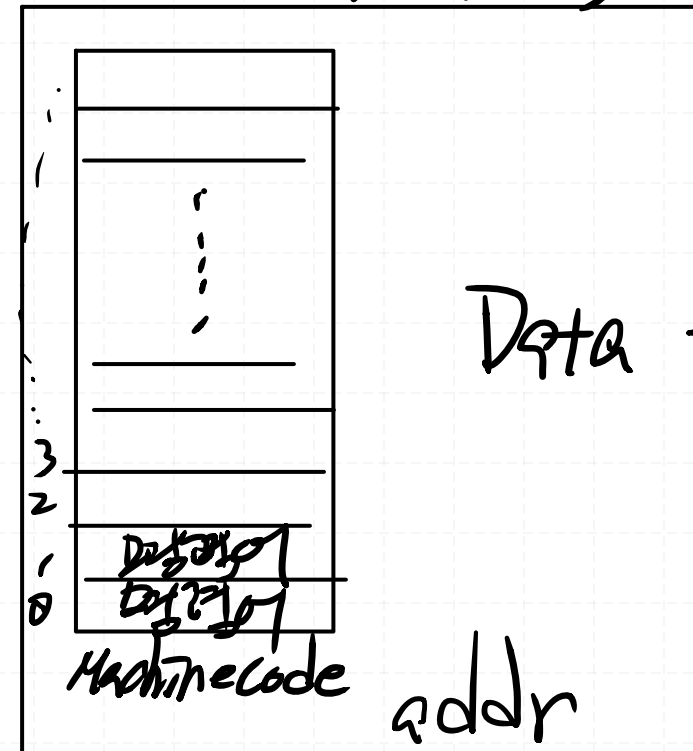


RISC-V (RV32I Instruction Set) 'R-Type' CPU (RISC-V, RV32I)

Instruction Memory (ROM/Flash)



Data

addr

Control Unit

OpCode
Func3
Func7

RFWe

aluControl

We

RA1

RA2

WA

WD

RD1

RD2

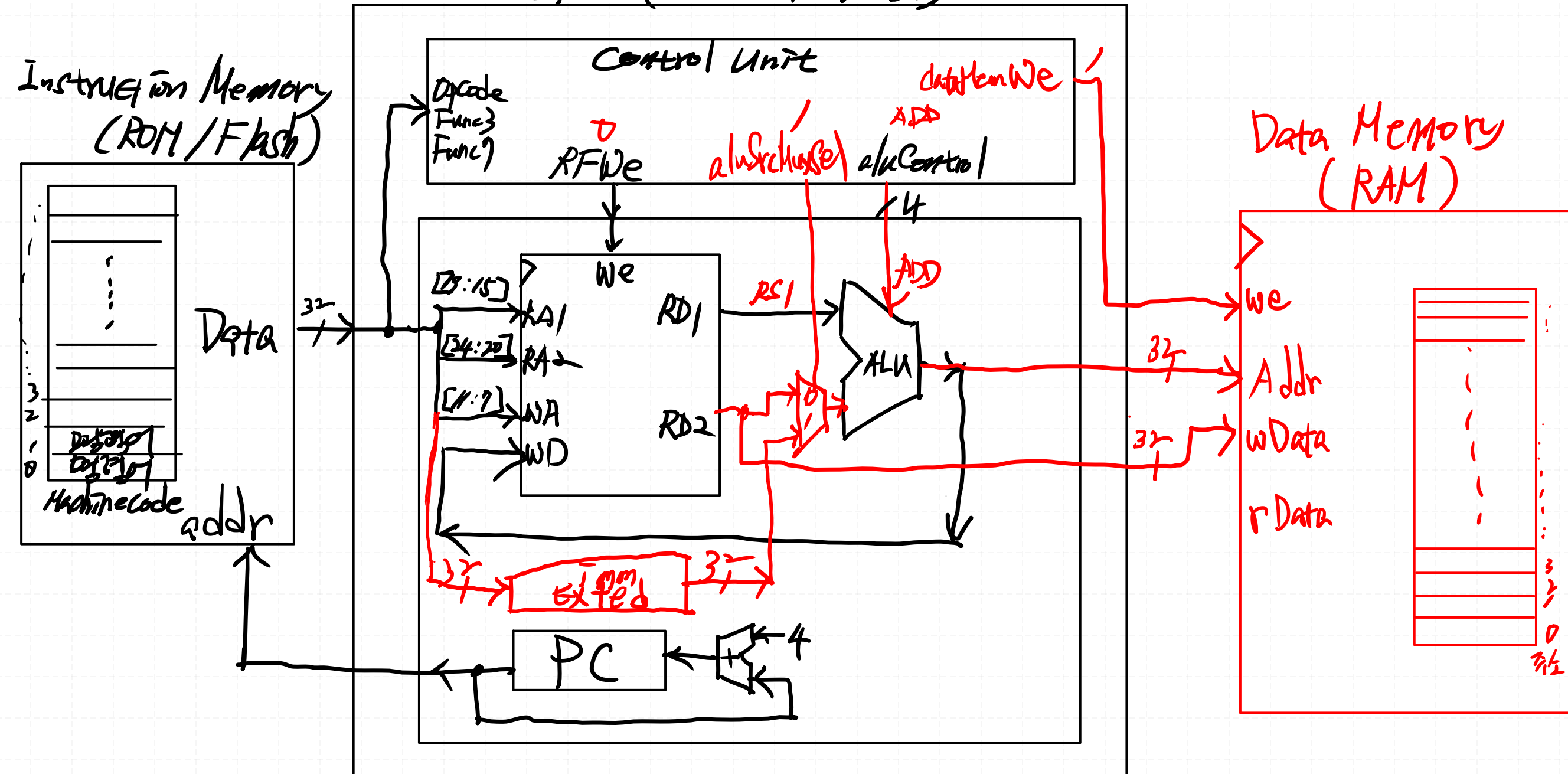
ALU

PC

WA

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	TYPE	MNEMONIC	NAME	Descript	Note
Funct7 (7)							Register Source 2 (5)					Register Source 1 (5)					Funct3 (3)			Register Destination (5)					Opcode (7)							R-TYPE				
0	0	0	0	0	0	0	rs2					rs1					0 0 0			rd					0 1 1 0 0 1 1								ADD	ADD	rd = rs1 + rs2	
0	1	0	0	0	0	0	rs2					rs1					0 0 0			rd					0 1 1 0 0 1 1								SUB	SUB	rd = rs1 - rs2	
0	0	0	0	0	0	0	rs2					rs1					0 0 1			rd					0 1 1 0 0 1 1								SLL	Shift Left Logical	rd = rs1 << rs2	
0	0	0	0	0	0	0	rs2					rs1					1 0 1			rd					0 1 1 0 0 1 1								SRL	Shift Right Logical	rd = rs1 >> rs2	
0	1	0	0	0	0	0	rs2					rs1					1 0 1			rd					0 1 1 0 0 1 1								SRA	Shift Right Arith*	rd = rs1 >>> rs2	msb-extends
0	0	0	0	0	0	0	rs2					rs1					0 1 0			rd					0 1 1 0 0 1 1								SLT	Set Less Than	rd = (rs1 < rs2) ? 1:0	
0	0	0	0	0	0	0	rs2					rs1					0 1 1			rd					0 1 1 0 0 1 1								SLTU	Set Less Than (U)	rd = (rs1 < rs2) ? 1:0	zero-extends
0	0	0	0	0	0	0	rs2					rs1					1 0 0			rd					0 1 1 0 0 1 1								XOR	XOR	rd = rs1 ^ rs2	
0	0	0	0	0	0	0	rs2					rs1					1 1 0			rd					0 1 1 0 0 1 1								OR	OR	rd = rs1 rs2	
0	0	0	0	0	0	0	rs2					rs1					1 1 1			rd					0 1 1 0 0 1 1							AND	AND	rd = rs1 & rs2		

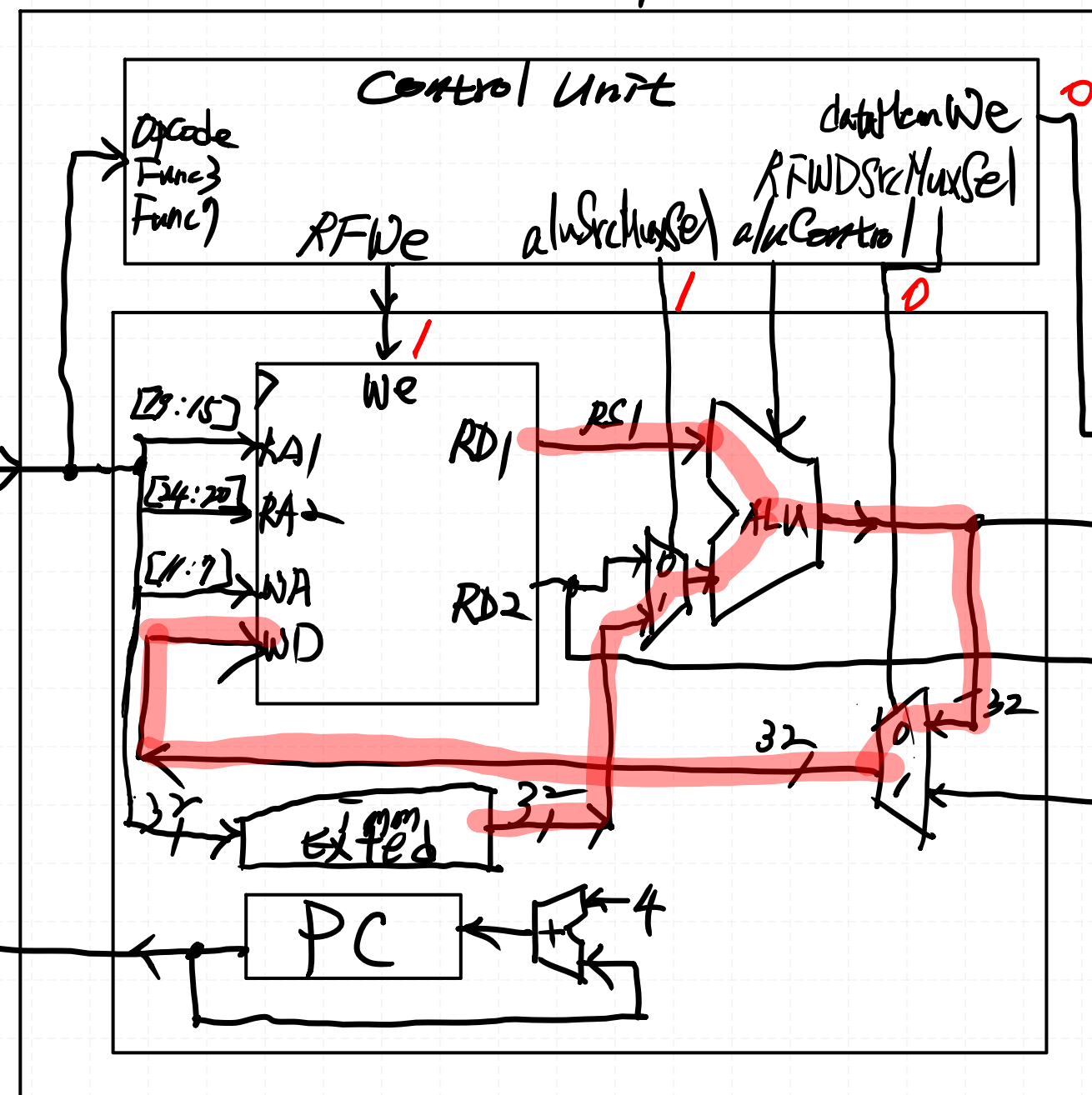
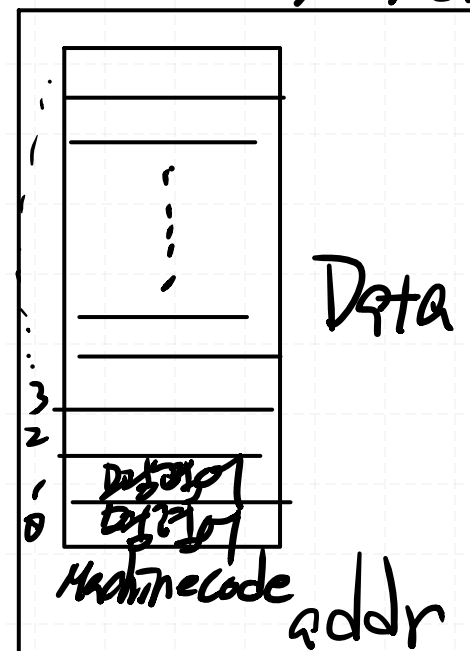
RISC-V (RV32I Instruction Set) 'S-Type' CPU (RISC-V, RV32I)



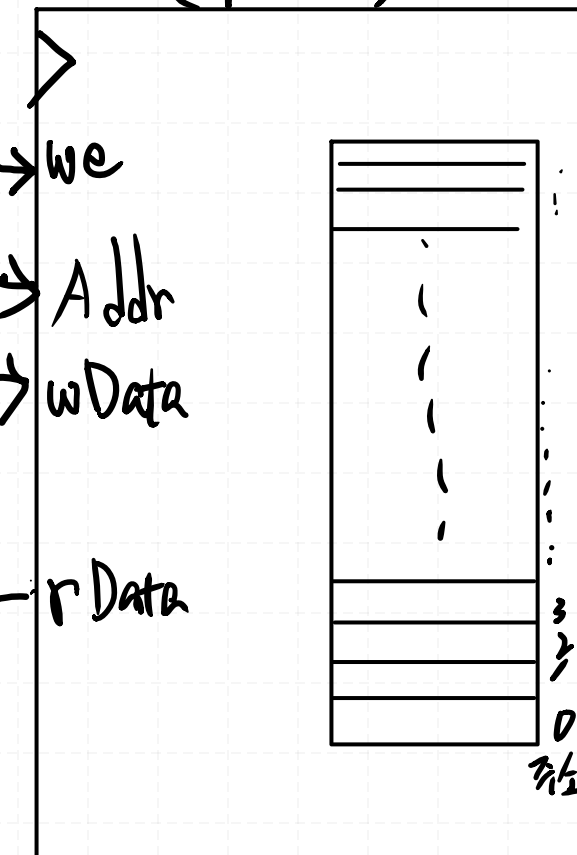
$M[rs1 + imm] = rs2$
Memory [31] = Data
 RAM

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	TYPE	MNEMONIC	NAME	Descript
immediate (7)							Register Source 2 (5)					Register Source 1 (5)					Funct3 (3)			immediate (5)					Opcode (7)										
imm[11:5]							rs2					rs1					0 0 0			imm[4:0]					0 1 0 0 0 1 1							store	SB	Store Byte 8	$M[rs1 + imm][0:7] = rs2[0:7]$
imm[11:5]							rs2					rs1					0 0 1			imm[4:0]					0 1 0 0 0 1 1							S-TYPE	SH	Store Half 16	$M[rs1 + imm][0:15] = rs2[0:15]$
imm[11:5]							rs2					rs1					0 1 0			imm[4:0]					0 1 0 0 0 1 1								SW	Store Word 32	$M[rs1 + imm][0:31] = rs2[0:31]$

Instruction Memory
(ROM/Flash)



Data Memory
(RAM)



$$rd = rs1 + imm$$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	TYPE	MNEMONIC	NAME	Descript
immediate (7)							shift amount (5)					Regitster Source 1 (5)					Funct3 (3)			Register Destination (5)					Opcode (7)										
imm[11:0]												rs1					0 0 0			rd					0 0 1 0 0 1 1							I-TYPE	ADDI	ADD Immediate	rd = rs1 + imm
imm[11:0]												rs1					0 1 0			rd					0 0 1 0 0 1 1								SLTI	Set Less Than Imm	rd = (rs1 < imm) ? 1 : 0
imm[11:0]												rs1					0 1 1			rd					0 0 1 0 0 1 1								SLTIU	Set Less Than Imm (U)	rd = (rs1 < imm) ? 1 : 0
imm[11:0]												rs1					1 0 0			rd					0 0 1 0 0 1 1								XORI	XOR Immediate	rd = rs1 ^ imm
imm[11:0]												rs1					1 1 0			rd					0 0 1 0 0 1 1								ORI	OR Immediate	rd = rs1 imm
imm[11:0]												rs1					1 1 1			rd					0 0 1 0 0 1 1								ANDI	AND Immediate	rd = rs1 & imm
0	0	0	0	0	0	0	shamt					rs1					0 0 1			rd					0 0 1 0 0 1 1								SLLI	Shift Left Logical Imm	rd = rs1 << shamt[0:4]
0	0	0	0	0	0	0	shamt					rs1					1 0 1			rd					0 0 1 0 0 1 1								SRLI	Shift Right Logical Imm	rd = rs1 >> shamt[0:4]
0	1	0	0	0	0	0	shamt					rs1					1 0 1			rd					0 0 1 0 0 1 1								SRAI	Shift Right Arith Imm	rd = rs1 >>> shamt[0:4]